Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “motor” and “generator” to compare your model or result. Sentence starters are optional.

motor: a device converting electrical energy to mechanical output

generator: a device converting mechanical input to electrical output

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A consistently drawn magnetic-field map has 3 times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives 20 J electrically and delivers 12 J mechanically; 8 J goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

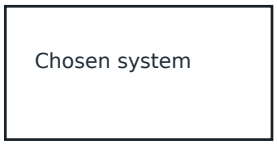
Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp.
Label each input and useful output.

Sentence starter: The … in my model represents … . Its label shows … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

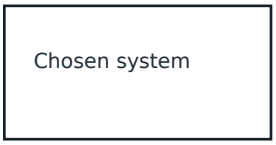
Marzano check - circle: 1 2 3

04 Apply the physics

A model-car motor receives 100 J of electrical energy and delivers 60 J of useful mechanical energy. Complete the flow model, calculate efficiency and identify the energy transferred to other forms.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A motor drives a generator that feeds the motor.
Explain why real losses prevent the loop from running forever without external input.

Sentence starter: My claim is … . The diagram/data show …, so … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

An electric field of 6 N/C points left.
Draw the force direction on +4 C and on -4 C, then compare sizes.
Draw field and force arrows separately for each charge.

Word help: electric field = force per positive test charge at a location

Sentence starter: Both … . They differ in … because … .

Field / first contribution

Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

07 Exit check

Which device converts mechanical input into electrical output?

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A motor draws the same electrical power in two designs, but one delivers more mechanical power.
Compare efficiencies and name a measurement needed.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3